Ising Model

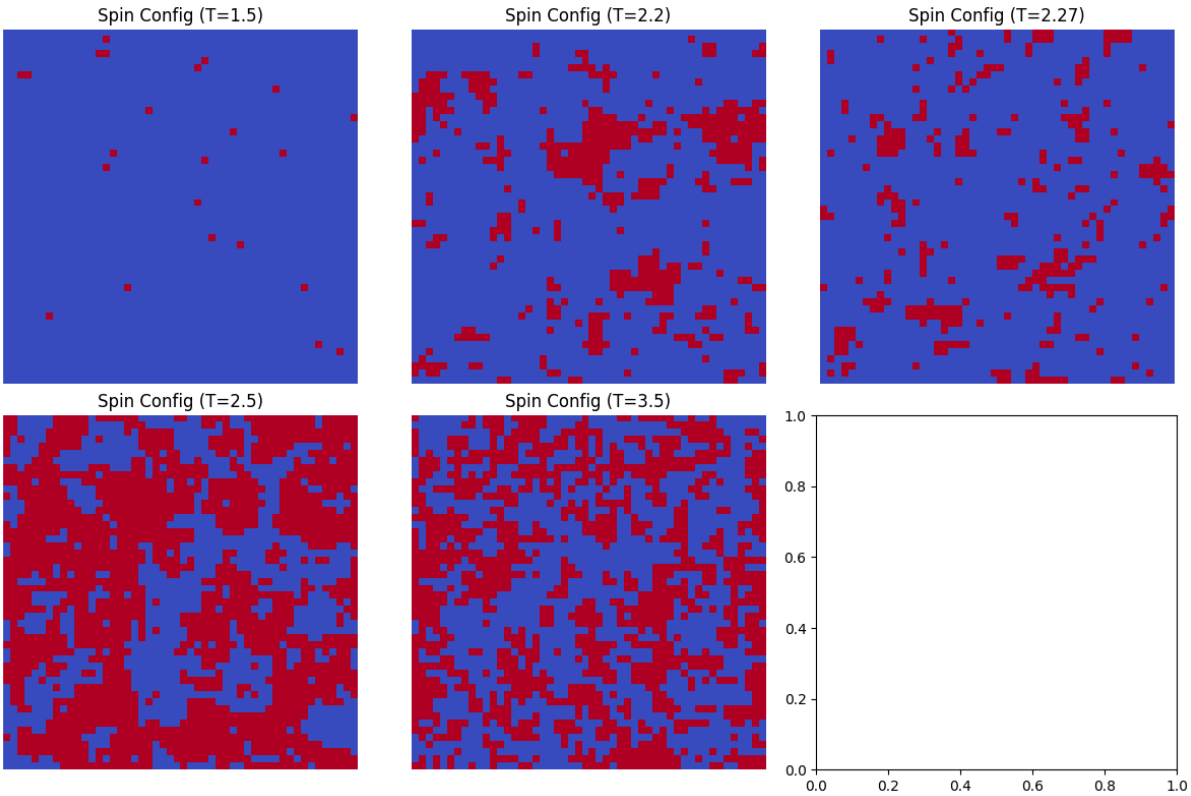
Class Assignment

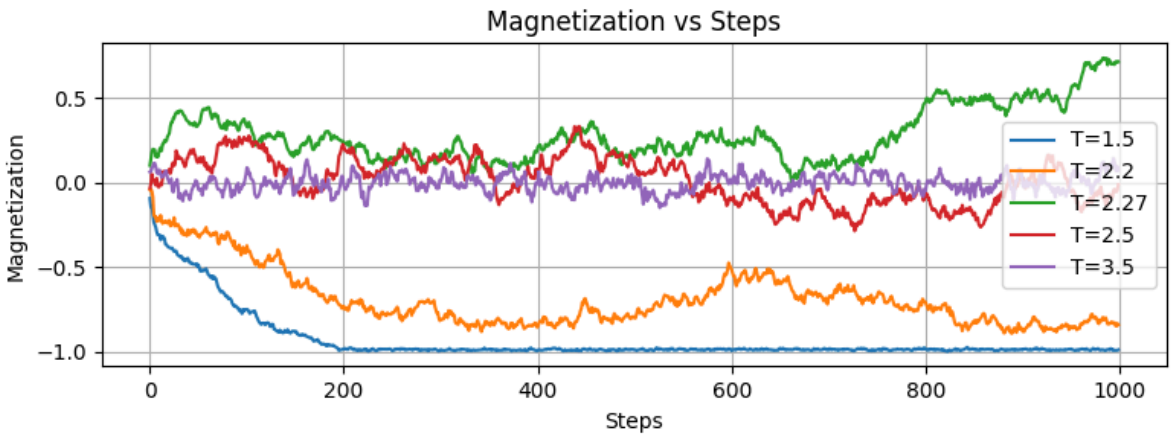
Sanjeev K
Ph25mscst11026

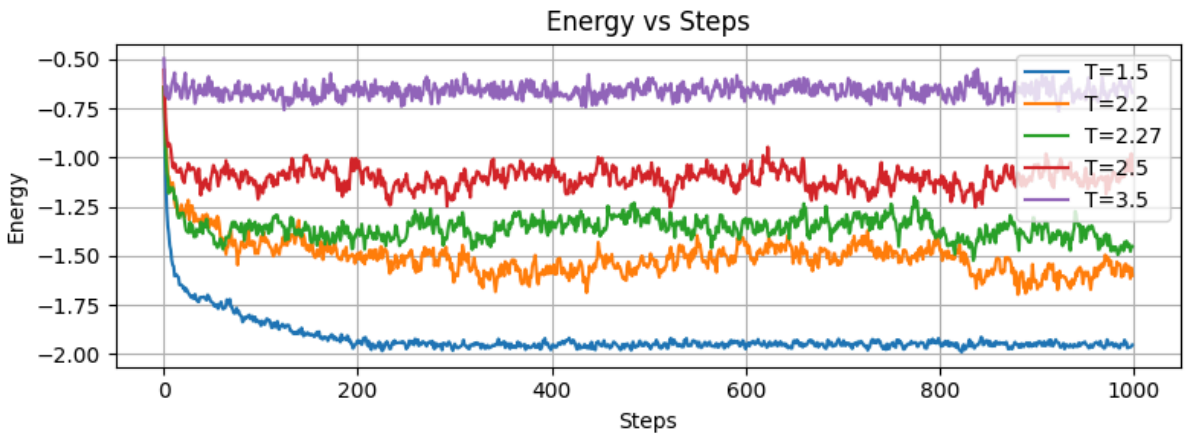
```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib import animation

# Parameters
L = 50                # Lattice size (LxL)
J = 1.0               # Interaction strength
n_steps = 1000        # Number of Monte Carlo steps

#5 representative temperatures
T_list = [1.5, 2.2, 2.27, 2.5, 3.5]


# Initialize lattice with spins (+1 or -1)
def initialize_lattice(L):
    return np.random.choice([-1, 1], size=(L, L))

# Calculate energy change for flipping spin at (i, j)
def delta_energy(L, J, lattice, i, j):
    spin = lattice[i, j]
    neighbors = lattice[(i+1)%L, j] + lattice[(i-1)%L, j] + \
                lattice[i, (j+1)%L] + lattice[i, (j-1)%L]
    return 2 * J * spin * neighbors

# Total energy of the system
def total_energy(L, J, lattice):
    E = 0
    for i in range(L):
        for j in range(L):
            spin = lattice[i, j]
            neighbors = lattice[(i+1)%L, j] + lattice[i, (j+1)%L]
            E -= J*spin * neighbors
    return E

# Perform Metropolis step
```

```python
def metropolis_step(L, J, lattice, T):
    for _ in range(L * L):
        i = np.random.randint(0, L)
        j = np.random.randint(0, L)
        dE = delta_energy(L, J, lattice, i, j)
        if dE < 0 or np.random.rand() < np.exp(-dE / T):
            lattice[i, j] *= -1

# Compute magnetization
def magnetization(L, lattice):
    return np.sum(lattice) / (L * L)

def simulate_ising_2d(L, J, T, n_steps):
    """Run the 2D Ising model simulation."""
    beta = 1.0 / T
    lattice = initialize_lattice(L)
    magnetizations = []
    energies = []

    for step in range(n_steps):
        metropolis_step(L, J, lattice, T)
        mag = magnetization(L, lattice)
        magnetizations.append(mag)
        en = total_energy(L, J, lattice) / (L * L)
        energies.append(en)

    return lattice, magnetizations, energies


fig, ax = plt.subplots(2, 3, figsize=(12, 8))
ax = ax.flatten()

for i, T in enumerate(T_list):
    final_spins, mags, Es = simulate_ising_2d(L, J, T, n_steps)

    # Plot spin configuration (MAIN REQUIRED)
    ax[i].imshow(final_spins, cmap='coolwarm')
    ax[i].set_title(f"Spin Config (T={T})")
    ax[i].axis('off')

plt.tight_layout()
plt.show()
```

```python
fig2, ax2 = plt.subplots(2, 1, figsize=(8,6))

for T in T_list:
    final_spins, mags, Es = simulate_ising_2d(L, J, T, n_steps)

    ax2[0].plot(mags, label=f"T={T}")
    ax2[1].plot(Es, label=f"T={T}")

ax2[0].set_title("Magnetization vs Steps")
ax2[0].set_xlabel("Steps")
ax2[0].set_ylabel("Magnetization")
ax2[0].legend()
ax2[0].grid()

ax2[1].set_title("Energy vs Steps")
ax2[1].set_xlabel("Steps")
ax2[1].set_ylabel("Energy")
ax2[1].legend()
ax2[1].grid()

plt.tight_layout()
plt.show()
```


Spin Config (T=1.5)
Spin Config (T=2.2)
Spin Config (T=2.27)
Spin Config (T=2.5)
Spin Config (T=3.5)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0


Magnetization vs Steps
Magnetization
0.5
0.0
−0.5
−1.0
0
200
400
600
800
1000
Steps
T=1.5
T=2.2
T=2.27
T=2.5
T=3.5


Energy vs Steps
Energy
−0.50
−0.75
−1.00
−1.25
−1.50
−1.75
−2.00
0
200
400
600
800
1000
Steps
T=1.5
T=2.2
T=2.27
T=2.5
T=3.5